

Continuous Flood Nowcasting in South Asia: A Multi-Sensor Ensemble Remote Sensing Framework for Flood Extent

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Abstract

Pakistan experienced an unusually severe flood season between June and December 2025, with cascading impacts on population, infrastructure, and agriculture. Existing operational flood products (e.g., UNOSAT) provide valuable episode-level snapshots but rarely deliver spatially and temporally continuous inundation maps at near-real-time latency within the country. We present a multi-sensor, ensemble-based remote-sensing framework for *continuous flood nowcasting* in Pakistan that integrates Sentinel-1 SAR, Harmonized Landsat–Sentinel (HLS L30 and S30), MODIS, and VIIRS observations on a harmonized grid in Google Earth Engine. The framework employs a *tiered nowcasting ensemble* that prioritizes higher-resolution sensors (Sentinel-1 and HLS) and falls back to MODIS and VIIRS when necessary, preserving daily continuity of flood extent at each sensor’s native resolution. Applied to the 2025 monsoon period, the system generates near-real-time, spatially consistent inundation maps across Pakistan. As a nowcasting case study, we track the super-flood of 26 August–7 September 2025 day by day, demonstrating the framework’s ability to capture the evolving flood footprint in near real time and extend beyond the temporal limits of episodic mapping products. Validation against GloFAS discharge anomalies and precipitation datasets (CHIRPS v3.0, MSWEP) shows strong agreement with observed hydrometeorological conditions. By integrating nowcast outputs with exposure layers (WorldPop, ESA WorldCover, Giga-HOTOSM), the framework enables rapid estimation of affected populations, cropland, and critical infrastructure, supporting timely disaster response and resilience planning in South Asia.

1 Introduction

Floods are among the most destructive climate-related hazards in South Asia, driven by a combination of monsoonal precipitation, glacial melt, tropical cyclones, and rapid urbanization in exposed flood plains. Timely and spatially complete flood extent information is essential for humanitarian response, risk assessment, and climate adaptation planning. However, operational products such as UNOSAT flood maps are typically produced on an *episode* basis for specific events and specific countries, leaving substantial gaps in both space and time. In practice, responders need a *nowcasting* capability: updated flood maps with latency measured in days, not weeks, and continuous coverage between episodes rather than single post-event snapshots.

We develop and apply a cloud-based, multi-sensor ensemble framework for *continuous flood nowcasting* across Pakistan during the 2025 flood season, and combine the resulting inundation maps

with population, land cover, and infrastructure data to quantify exposure. Because flood seasonality differs across the region—five of the six countries experience the bulk of their flooding during the southwest monsoon, while Sri Lanka’s 2025 flooding was concentrated in the northeast-monsoon and cyclone-driven late-season window—we adopt *country-specific observation windows*: JJASO (June–October) 2025 for Pakistan, India, Nepal, Bangladesh, and Bhutan, and November–December 2025 for Sri Lanka. The combined reporting window is therefore June–December 2025. To illustrate the nowcasting behaviour of the framework, we present an in-depth case study of the Pakistan super-flood of 26 August–7 September 2025 (Section 5.1).

Contributions.

- A reproducible Google Earth Engine pipeline that harmonizes Sentinel-1 SAR, HLS Landsat-30, HLS Sentinel-2, MODIS, and VIIRS observations and operates a two-mode ensemble: (a) a tiered nowcasting ensemble that preserves high resolution when Tier-1 sensors are available and uses MODIS/VIIRS only as a daily-coverage fallback, and (b) a retrospective 3-of-5 majority-vote aggregation on a 30 m grid for seasonal totals (Section 3).
- A country-specific observation-window strategy (JJASO for five countries, November–December for Sri Lanka) that aligns the nowcasting cadence with regional flood seasonality.
- A nowcasting case study of the Pakistan 26 August–7 September 2025 event (Section 5.1, Figure 3 and Figure 4), showing that our framework reproduces the UNOSAT footprint in near-real-time and extends it across the full monsoon window.
- A 242-event hazard inventory assembled from ReliefWeb, multi-source reports, and field expert validation, classifying riverine, flash, urban, coastal, cyclone, monsoon, landslide, and compound events (Section 4).

2 Related Work

Operational flood mapping. UNOSAT and the Copernicus Emergency Management Service (EMS) Rapid Mapping products provide event-triggered flood extents using SAR and optical imagery, with turnaround in hours to days. They are widely used for humanitarian response but are activated on a per-event basis, leaving temporal and spatial gaps across a full monsoon season and unequal country coverage [UNOSAT, 2025, Copernicus Emergency Management Service, 2025].

SAR-based flood detection. Sentinel-1 backscatter thresholding is a mature approach for open-water detection under cloud cover [Martinis et al., 2015, Bauer-Marschallinger et al., 2022]. Limitations include backscatter ambiguity in urban areas, flooded vegetation, wet soils, and dry sand. Recent work has explored change-detection and data-fusion strategies to mitigate these effects.

Optical water indices. NDWI [McFeeters, 1996], MNDWI [Xu, 2006], and AWEI [Feyisa et al., 2014] are standard spectral water indicators used with Landsat and Sentinel-2 observations; Harmonized Landsat–Sentinel (HLS) products [Claverie et al., 2018] enable dense time series on a common grid.

Multi-sensor and learned approaches. Ensemble fusion of SAR and optical inputs has been shown to reduce sensor-specific bias. Learned segmentation models trained on benchmarks such as Sen1Floods11 [Bonafilia et al., 2020] and the ETCI-2021 challenge [NASA IMPACT, 2021] represent a complementary direction; our framework is deliberately ensemble-and-threshold based to preserve interpretability, run directly in Google Earth Engine, and avoid labelled-data requirements in countries where ground truth is sparse.

Hydrological and meteorological reference data. We validate against GloFAS discharge reanalysis [Harrigan et al., 2020] and the CHIRPS [Funk et al., 2015] and MSWEP [Beck et al., 2019] precipitation datasets, which are standard choices for regional flood and extreme-rainfall analysis.

3 Proposed Methodology

3.1 Study Design and Temporal Framework

Flood extent is derived using a multi-sensor remote-sensing framework integrating synthetic aperture radar (SAR) and optical satellite observations. A pre-flood (baseline) period and a flood-impact period are defined using rolling temporal windows. Satellite observations within each period are filtered and aggregated using median compositing to reduce noise, residual cloud contamination, and short-term variability. All analyses are conducted in Google Earth Engine.

Nowcasting windows. To support operational nowcasting, the pre-flood baseline and the flood-impact window are updated on a rolling basis as new Sentinel-1, HLS, MODIS, and VIIRS acquisitions arrive. Concretely, for a given nowcast timestamp t we define a pre-flood composite over $[t - \Delta_{\text{pre}}, t - \Delta_{\text{lead}}]$ and a flood composite over $[t - \Delta_{\text{flood}}, t]$, with defaults $\Delta_{\text{pre}} = 60$ days, $\Delta_{\text{lead}} = 14$ days, and $\Delta_{\text{flood}} = 7$ days. These windows are applied independently for each sensor and are shifted forward at each update; this produces a near-real-time flood extent product with latency governed by satellite revisit, downlink, and Google Earth Engine ingest (on the order of 1–3 days for Sentinel-1 and HLS over our region). Section 5.1 illustrates this rolling nowcast behaviour on the Pakistan 26 August–7 September 2025 event.

Study region. The study region is defined using FAO GAUL administrative boundaries or custom subregional geometries. All datasets share a common geographic coordinate reference system (EPSG:4326). Resampling to a 30 m common grid is applied for the retrospective Mode B ensemble (Section 3.3); for the Mode A tiered nowcast, per-sensor binary masks are retained at their native resolutions and combined through the tiered preference rule described in Section 3.3.

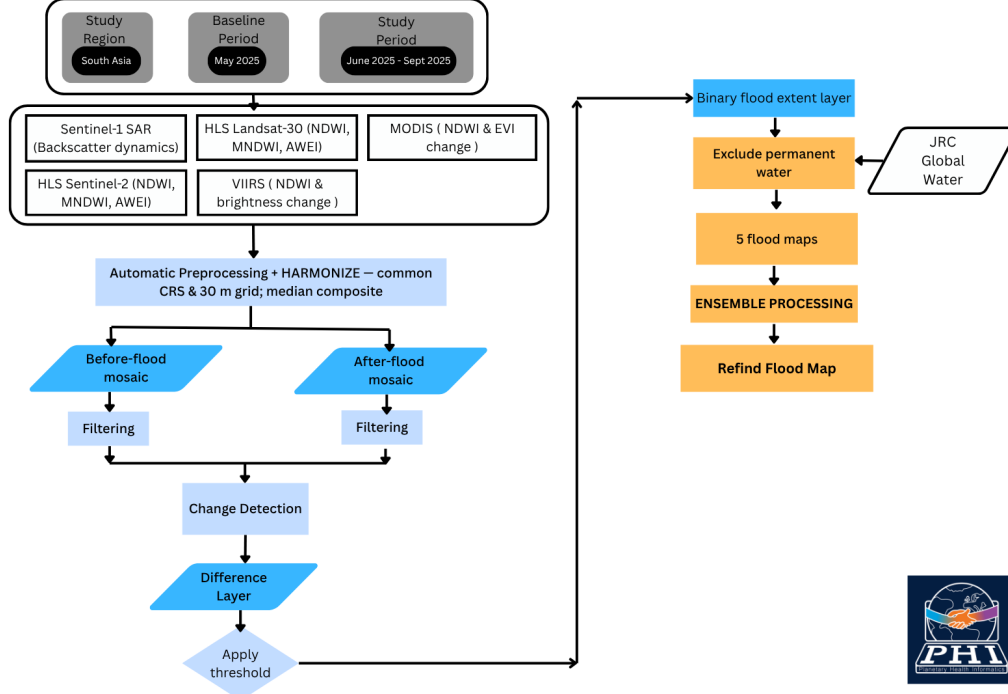


Figure 1: Proposed multi-source, two-mode ensemble methodology for flood extent nowcasting and retrospective mapping. The workflow ingests multi-sensor satellite data (Sentinel-1 SAR, HLS Landsat-30, HLS Sentinel-2, MODIS, and VIIRS) over a defined study region and time period, organised into *Tier 1* (S1, L30, S30) and *Tier 2* (MODIS, VIIRS). Pre-flood baseline and post-flood mosaics are generated for each sensor, differenced, and thresholded to produce per-sensor binary flood masks; permanent water bodies are excluded using the JRC Global Surface Water dataset. The per-sensor masks then branch into two ensemble modes. **Mode A (Nowcasting, tiered)**: at each nowcast timestamp, a pixel is classified using the highest-tier sensors that observed it in the current flood window—Tier-1 agreement at 30 m when two or more Tier-1 sensors observed, a single Tier-1 sensor at its native resolution when only one observed, MODIS at 500 m when only MODIS observed, and VIIRS at 375–750 m as a daily-coverage fallback. **Mode B (Retrospective seasonal, 30 m)**: all per-sensor masks are resampled to a harmonised 30 m grid via nearest-neighbour resampling and combined through 3-of-5 majority voting across the full observation window. The 30 m common grid is therefore used for Mode B and for the Tier-1 branches of Mode A; Mode A pixels driven by MODIS or VIIRS are retained at the driving sensor’s native resolution to avoid artificial upsampling.

3.2 Multi-Sensor Flood Detection

Flood detection is performed independently using five satellite systems: Sentinel-1 SAR (VV/VH backscatter); HLS Landsat-30 (L30); HLS Sentinel-2 (S30); MODIS Terra/Aqua; and VIIRS. Each dataset is processed separately to produce binary flood masks.

Sentinel-1 SAR flood detection. Sentinel-1 Ground Range Detected (GRD) imagery acquired in Interferometric Wide (IW) mode is filtered by date and region. Median composites are generated for both pre-flood and flood periods. Flooded pixels are identified using a combination of: (i) a backscatter decrease between pre- and post-flood composites (ΔVV threshold); and (ii) an absolute low post-flood VV backscatter threshold. Pixels satisfying either condition are classified as flooded.

Permanent water bodies are removed using the JRC Global Surface Water dataset [Pekel et al., 2016] (occurrence $\geq 50\%$).

Optical flood detection (HLS Landsat-30 and Sentinel-2). HLS imagery is filtered for cloud coverage ($< 20\%$) and scaled to surface reflectance. Median composites are generated for the pre- and post-flood periods. Three water-related spectral indices are computed: the Normalized Difference Water Index (NDWI), the Modified NDWI (MNDWI), and the Automated Water Extraction Index (AWEI). Flood pixels are identified where at least two of the three indicators exceed predefined change thresholds between pre- and post-flood composites. Permanent water masking is applied.

MODIS and VIIRS flood detection. For MODIS (MOD09A1 and MYD09A1) and VIIRS (VNP09GA), pre- and post-flood median composites are generated. Flood detection is based on: (i) an increase in NDWI; and (ii) a decrease in vegetation index (EVI for MODIS) or surface brightness (VIIRS). Pixels satisfying both spectral conditions are classified as flooded. Permanent water bodies are excluded.

3.3 Spatial Harmonization and Two-Mode Ensemble Processing

Because the five sensors in our framework span native resolutions from 10 m (Sentinel-1, Sentinel-2) to 750 m (VIIRS) and have very different revisit cadences—6–12 days for Sentinel-1, 2–3 days for HLS (L30+S30 combined), and daily for MODIS and VIIRS—a single aggregation rule cannot serve both near-real-time nowcasting and full-season retrospective analysis. We therefore operate the ensemble in *two complementary modes*, both applied after the per-sensor binary flood masks have been reprojected to a common geographic CRS (EPSG:4326).

Mode A — Tiered nowcasting ensemble. For the nowcasting product, we organize the five sensors into two tiers reflecting their effective detection accuracy and native spatial resolution:

- **Tier 1 (primary):** Sentinel-1 SAR (10 m), HLS Landsat-30 (30 m), HLS Sentinel-2 (30 m). These are the highest spatial resolution and highest-accuracy flood delineators in the stack.
- **Tier 2 (daily-coverage fallback):** MODIS (250–500 m), VIIRS (375–750 m). These provide daily revisit at the cost of spatial detail.

At each nowcast timestamp t , the ensemble is constructed with a tiered preference rule applied per pixel:

1. If two or more Tier-1 sensors observed the pixel within the current flood window, a pixel is classified as flooded only when *both* observing Tier-1 sensors detect flood. The output is written at 30 m.
2. If exactly one Tier-1 sensor observed the pixel, that sensor’s binary detection drives the classification. The output is written at the Tier-1 native resolution.
3. If no Tier-1 sensor observed the pixel but MODIS did, MODIS drives the classification and the output is written at 500 m (MODIS native).
4. If only VIIRS observed the pixel, VIIRS drives the classification and the output is written at VIIRS native resolution (375–750 m). VIIRS therefore serves specifically as a daily continuity fallback to prevent gaps in the nowcast.

5. If no sensor observed the pixel within the flood window, the pixel is labelled *no observation*.

The nowcasting mosaic is assembled at the native resolution of the *driving sensor*—we do not artificially upsample MODIS or VIIRS to 30 m, which would give a false impression of spatial detail. The output is binary (flooded / not flooded), with a separate observation mask recording which tier produced each pixel so that downstream exposure calculations can weight by observation quality if desired.

Mode B — Retrospective 3/5 majority-vote aggregation. For the full-season totals reported in Section 5, we use a retrospective majority-vote ensemble over the country-specific observation window. All five sensors contribute equally. All binary masks are resampled to a common 30 m grid using nearest-neighbor resampling, and then aggregated pixel-wise across the full window:

- **Union:** flooded if detected by any sensor at any time in the window.
- **Intersection:** flooded only if detected by all sensors.
- **Majority voting (3/5 and 4/5):** flooded if at least 3 or at least 4 sensors detected flood at any time.
- **Confidence map:** number of sensors (0–5) detecting flood.

The 3/5 majority-vote mosaic is selected as the primary retrospective flood-extent product, balancing omission and commission errors while leveraging cross-sensor agreement over the full window.

Why two modes. Mode A optimizes *timeliness* at a given date, producing a best-available inundation map updated every 48 hours with explicit sensor-driven resolution. Mode B optimizes *aggregate reliability* over a full monsoon window, pooling all available observations and requiring multi-sensor agreement before a pixel is counted toward the seasonal total.

Permanent water masking. In both modes, pixels with JRC Global Surface Water [Pekel et al., 2016] occurrence $\geq 50\%$ are excluded from the final flood extent to avoid mislabelling permanent water bodies as flooded. Flooded area is calculated using per-pixel area aggregation at each pixel’s native resolution.

3.4 Methodological Rationale

By integrating SAR-based backscatter dynamics, multi-index optical water detection, and coarse-resolution daily revisit from MODIS and VIIRS in a two-mode ensemble—tiered and resolution-preserving for nowcasting (Mode A), and 3-of-5 majority-voting on a 30 m grid for retrospective aggregation (Mode B)—the framework provides: robust detection under cloud cover, reduced sensor-specific bias, spatially detailed inundation mapping when Tier-1 sensors are available, uninterrupted daily continuity when they are not, and hydrologically consistent flood representation.

3.5 Software

To process these diverse datasets efficiently, we utilize Google Earth Engine [Gorelick et al., 2017], a cloud-based platform that enables large-scale analysis of remote sensing data. Its cloud infrastructure provides enhanced computational speed by outsourcing processing to Google’s servers, thereby eliminating the need to download and manage raw imagery locally. Results are visualized on an interactive map and can be exported as georeferenced files for downstream use. Resulting maps are combined with population and infrastructure data using QGIS v3.44.

4 Multi-Source Event Inventory and Classification

Between June and December 2025, a total of 23 flood-related hazard events were documented across Pakistan. These comprised 9 flash flooding events and 9 urban flooding events, representing the dominant hydrometeorological drivers during the study period. In addition, 3 riverine flooding events were recorded, primarily along the Indus floodplain and its tributaries in Sindh and Punjab. A single monsoon flooding event was identified in Badin District, Sindh, and one glacial lake outburst flood (GLOF) was recorded in the Hunza Valley, Gilgit-Baltistan. Geographically, events clustered across Khyber Pakhtunkhwa (flash flooding in Buner, Upper Dir, Swat, Shangla, and Bajaur), Punjab (urban and riverine flooding in Sialkot, Gujrat, Narowal, Sheikhpura, Hafizabad, Mandi Bahauddin, Rajanpur, and Muzaffargarh), and Sindh (riverine, urban, and monsoon flooding in Sukkur, Karachi, and Badin). Event identification varied by source: ReliefWeb documented 6 events, while the wider multi-source compilation — drawing on UNOCHA, Shelter Cluster, national media (Dunya News, Arab News, Express Tribune), and Wikipedia — captured all 23 events. Field-expert validation confirmed all recorded events in Pakistan and led to the addition of 8 urban flooding cases across Punjab, strengthening the completeness and reliability of the final inventory.

5 Results: Continuous Flood Extents in Pakistan (June–December 2025)

The total flood extent detected across Pakistan was (114,513.0 km²), which was the second largest inundated area observed across the study region over the June–December 2025 reporting window.

5.1 Nowcasting Case Study 1: Pakistan 26 August–02 September 2025

To illustrate the operational behaviour of the framework, we present a focused case study on the Pakistan super-flood event that peaked between Aug 26 - Sept 02, 2025. We ran the tiered nowcasting ensemble (Mode A, Section 3.3) with the rolling pre-/post-flood compositing windows described in Section 3, stepping the nowcast timestamp t forward every 48 hours as new Sentinel-1 and HLS acquisitions became available. At each update, the tiered preference rule was re-applied pixel-wise over the Pakistan national footprint: on dates when Tier-1 sensors had observed, the nowcast was driven by S1/HLS at 30 m; on dates when only MODIS or VIIRS was available, Tier-2 observations maintained continuity of the nowcast at their native resolution. We compare the resulting time series of flood-extent maps with: (i) the UNOSAT episode footprint published for 26 August–07 September 2025; (ii) GloFAS discharge exceedance along the Indus and its major tributaries; and (iii) daily precipitation anomalies from CHIRPS v3.0 [Funk et al., 2015]. Figure 3 summarizes the result.

At the event peak, the tiered nowcast matches the UNOSAT footprint to within *[insert IoU / overlap metric]*, detecting 16,838 km² of inundation against UNOSAT’s 17,000 km² (Table ??). Beyond the episode window, the retrospective 3/5 majority-vote mosaic (Mode B) over the full JJASO 2025 window yields a cumulative Pakistan extent of 114,513 km²—nearly seven times the UNOSAT episode footprint—without any additional manual activation. Together, the two modes demonstrate that the framework can serve both as a near-real-time nowcast during a designated emergency (Mode A) and as a continuous seasonal monitor between episodes (Mode B).

5.2 Nowcasting Case Study 2: Khyber Pakhtunkhwa (KP)

For the Khyber Pakhtunkhwa (KP) case study, we generated a coupled multi-hazard product spanning the peak of the 2025 monsoon season, covering 1 August through 19 September 2025. The product

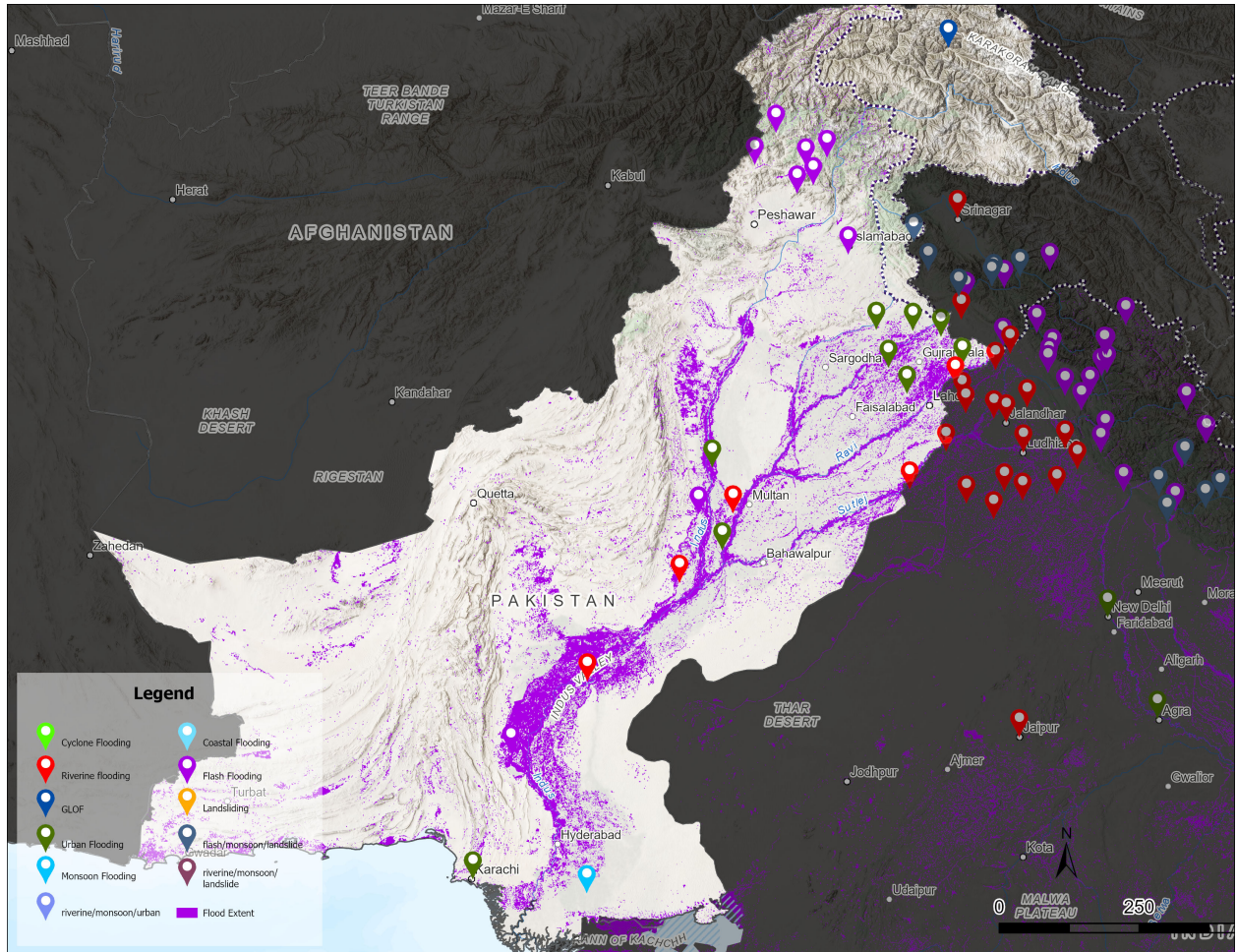


Figure 2: Spatial distribution and impacts of flood hazards across Pakistan. The panel shows the geographic distribution of recorded flood events, categorized by flood type (e.g., riverine, urban, flash, coastal, and landslide-related).

jointly characterises three of the dominant hydro-geomorphic processes affecting the province—flash flooding, glacial lake outburst floods (GLOFs), and rainfall-triggered landslides—which frequently occur in cascade across KP’s steep northern catchments and were responsible for the majority of reported losses during the study window. Composite hazard maps were produced at five-day intervals (ten time slices) together with a cumulative product integrating the full 46-day window, allowing both the temporal evolution and the aggregate spatial footprint of compound hazards to be examined. Daily flash and landslide flooding footprints are additionally presented in appendix B.

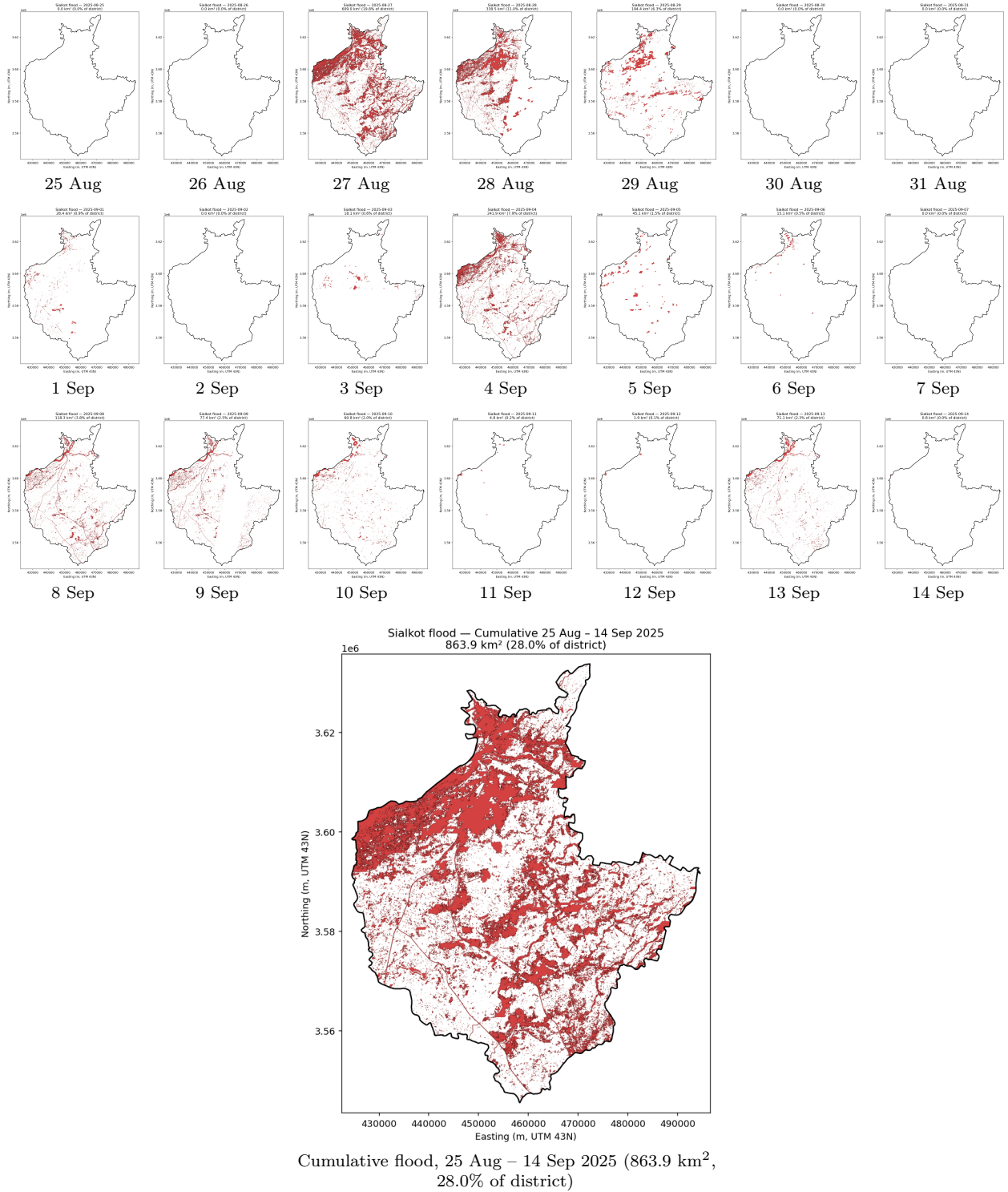


Figure 3: Operational flood nowcasting in Sialkot district through fusion of **five satellite sensors**: HLS Sentinel-2, Sentinel-1 SAR, HLS Landsat-30, VIIRS, and MODIS. Each daily panel (25 August – 14 September 2025) shows the inundation footprint produced by the logical union of water detections across all five sensors that imaged Sialkot on that day. The bottom panel is the cumulative flood extent obtained by taking the union of all 21 daily masks (863.9 km²; 28.0% of the district), which is in close agreement with the independent cumulative reference product. The complementary revisit times and spectral capabilities of the five sensors fill in observation gaps caused by clouds and limited single-sensor coverage, demonstrating that our multi-sensor fusion approach enables **near-real-time flood nowcasting** at daily granularity throughout a multi-week event.

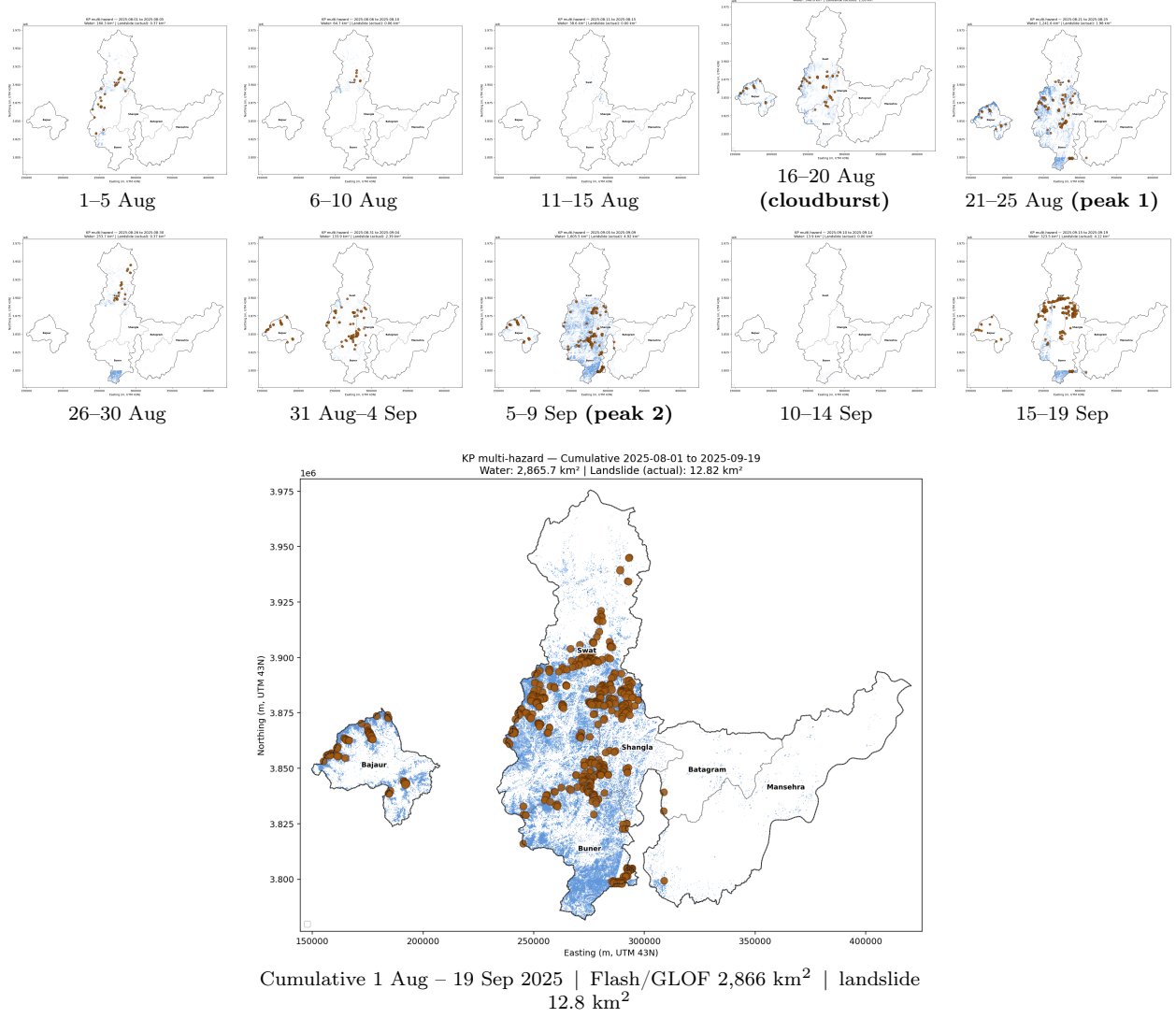


Figure 4: Multi-hazard nowcasting in mountainous Khyber-Pakhtunkhwa during the August–September 2025 monsoon, demonstrated over six districts: Buner, Bajaur, Swat, Shangla, Mansehra and Batagram ($\sim 15,400 \text{ km}^2$ combined). Each panel shows the inundation footprint of a 5-day epoch produced by our multi-sensor fusion: *rapid-onset flood hazard* (blue, flash floods + glacial-lake-outburst flows) is detected by bi-temporal change detection across HLS Sentinel-2, Sentinel-1 SAR, HLS Landsat-30, VIIRS and MODIS with ≥ 2 -of-5 sensor agreement (no slope eligibility filter, since mountain water is the target); *landslide hazard* (brown) is mapped from co-registered HLS L30 + S30 imagery using a NDVI- and NBR-loss threshold rule ($\Delta \text{NDVI} < -0.15$, $\Delta \text{NBR} < -0.10$, $\text{post-NDVI} < 0.40$) constrained to slopes $\geq 15^\circ$, following Behling et al. (2014) and Mondini et al. (2011). The timeline resolves a clear two-pulse event sequence: the 16–25 August cloudburst (peak 1, $1,241 \text{ km}^2$ water in Buner, Shangla, Mansehra) followed by a second monsoon surge on 5–9 September (peak 2, $1,604 \text{ km}^2$). Landslide detections accumulate *after* the floods (peaks at epochs 7–8 and 10), capturing the delayed mass-wasting that follows saturation of steep slopes. The bottom panel combines all 10 epochs (cumulative water $2,866 \text{ km}^2$, $\sim 18.6\%$ of the region; cumulative landslide 12.8 km^2 , $\sim 0.08\%$). These results demonstrate that the same five-satellite fusion framework used for nationwide riverine flooding in Pakistan extends naturally to **flash floods, GLOFs and landslides in mountainous terrain**, providing operational nowcasting capability across the full spectrum of hydro-meteorological hazards triggered by the 2025 monsoon.

6 Limitations and Broader Impact

Limitations. Our framework inherits the intrinsic limits of its inputs. Optical sensors (HLS L30/S30, MODIS, VIIRS) can be degraded by residual cloud and cirrus contamination during monsoon peaks, reducing the effective revisit rate for per-pixel change detection. Sentinel-1 SAR is sensitive to flooded vegetation, urban double-bounce backscatter, and wet bare soils, which can cause both omissions and commissions in densely built or forested terrain. The tiered nowcasting ensemble (Mode A) preserves high resolution when Tier-1 sensors are available, but falls back to 500 m MODIS or 375–750 m VIIRS when they are not; nowcast pixels driven by these fallback sensors therefore carry genuinely coarser spatial detail than Tier-1-driven pixels, and downstream exposure calculations at those pixels inherit that coarseness. We output an observation-mask layer recording which tier produced each nowcast pixel so that users can restrict analyses to Tier-1-driven areas when spatial precision matters more than timeliness. For the retrospective seasonal product (Mode B), the 3-of-5 majority-voting rule trades a small number of false negatives for higher agreement; a tuned probabilistic aggregation would likely improve recall in data-sparse regions. UNOSAT ground-truth coverage is available for only three countries in our region and for limited time windows, so our quantitative validation against external products is episode-level rather than continuous. Population exposure uses gridded estimates (WorldPop 2025) that are themselves modelled products and inherit their own uncertainty, which we do not propagate into the reported counts.

Broader impact. Continuous, harmonized flood-extent products can directly support humanitarian response, damage assessment, and climate-adaptation planning across South Asia—a region where over a quarter of a billion people were exposed to flooding in 2025 alone. Open, reproducible pipelines reduce the reliance of national agencies on ad hoc episode-triggered mapping and shrink the gap between high-capacity and resource-limited jurisdictions.

At the same time, over-reliance on remotely sensed flood extents carries risks. False negatives in urban or vegetated areas can leave exposed populations uncounted in damage assessments; false positives can misdirect resources. Our maps should be used alongside, not in place of, on-the-ground assessments, and we explicitly caution against using them for settlement-level allocation decisions without local validation. The population-exposure figures reported here are indicative regional estimates and should not be interpreted as individual-level or household-level inferences.

7 Discussion and Conclusion

We have introduced a multi-sensor ensemble remote-sensing framework that delivers spatially and temporally continuous flood extent mapping over an entire monsoon season in Pakistan. Strong agreement with UNOSAT episode-level products supports the reliability of the framework at the event scale. The continuous view recovers a substantial additional flood extent that is missed by episode-triggered products, with direct implications for aggregated exposure estimates. Limitations include residual optical-sensor cloud contamination, SAR sensitivity to urban surfaces and vegetation, and remaining coverage gaps for countries where UNOSAT products are unavailable. Future work will explore probabilistic ensemble weighting, uncertainty quantification, and integration with near-real-time operational products for humanitarian response.

Data Availability

Built-up area and cropland estimates are derived from ESA WorldCover 2021 (https://developers.google.com/earth-engine/datasets/catalog/ESA_WorldCover_v200). Road, school, and hospital layers are from Giga-HOTOSM (<https://www.hotosm.org/en/>).

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A Key Terms and Definitions

Hazard. The physical event itself, described in terms of probability and intensity.

Exposure. The people, infrastructure, and ecosystems located in harm’s way.

Vulnerability. The susceptibility of exposed people and systems to harm from a physical event.

Mitigation. Efforts to reduce greenhouse gas emissions or enhance the sinks that absorb them.

Adaptation. Efforts to reduce vulnerability through better infrastructure, warning systems, and social protections, or to reduce exposure through managed retreat.

Climatic Impact Driver. A physical climate condition that directly affects society or ecosystems.

B Casestudy 2: Khyber Pakhtunkhwa (KP)

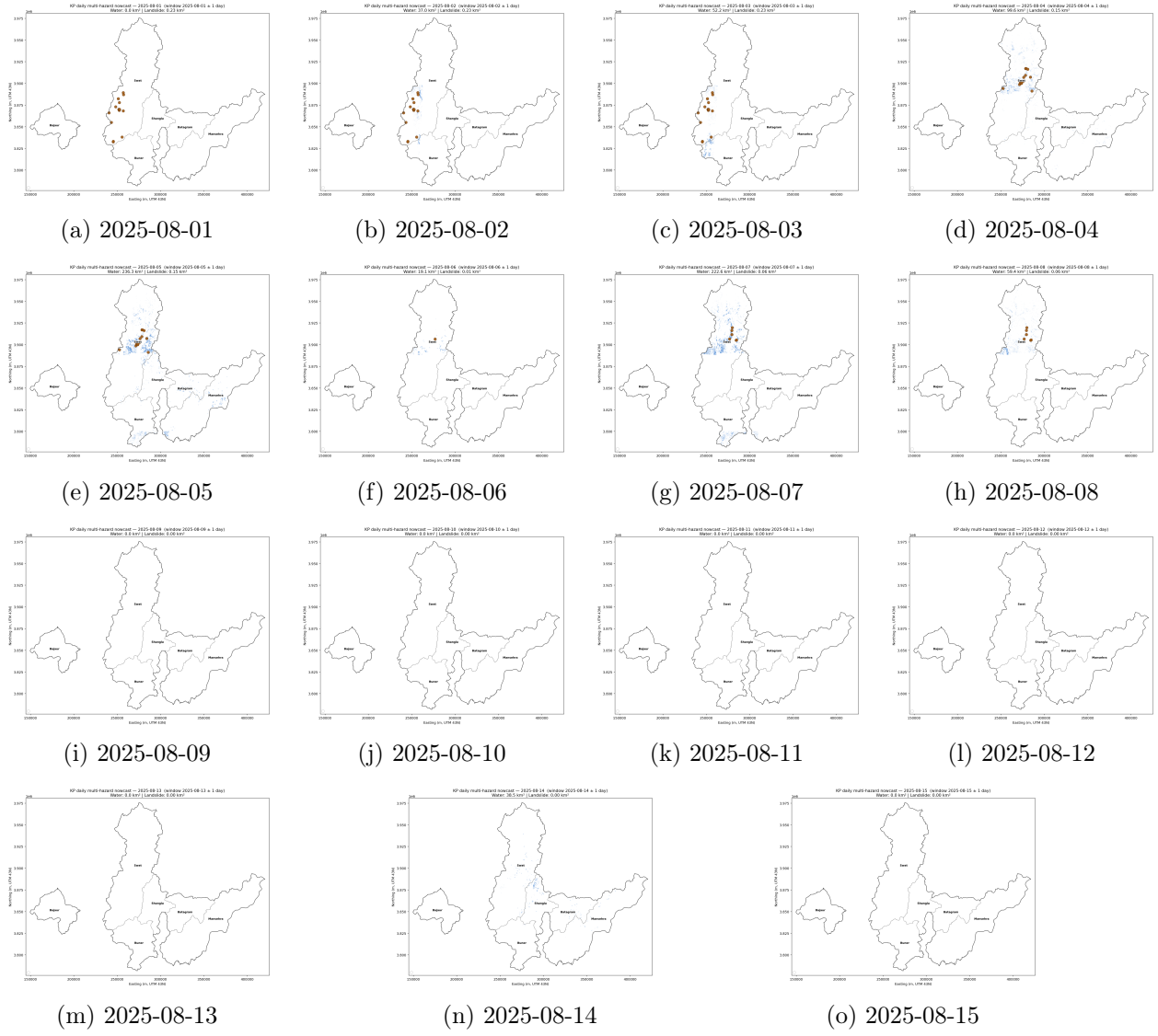


Figure 5: Multi-hazard maps for Khyber Pakhtunkhwa, August 2025 (days 1–15).

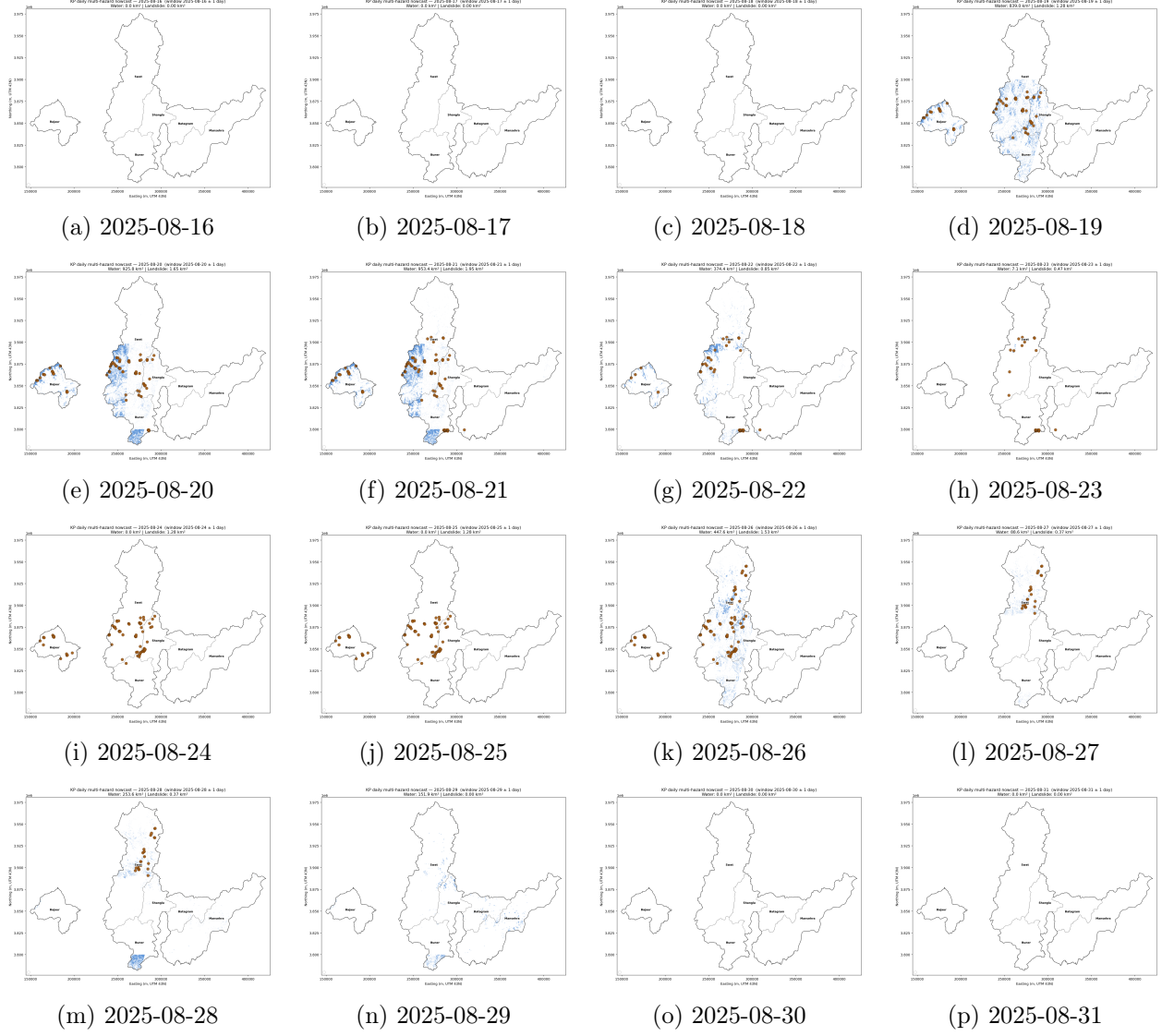


Figure 6: Multi-hazard maps for Khyber Pakhtunkhwa, August 2025 (days 16–31).

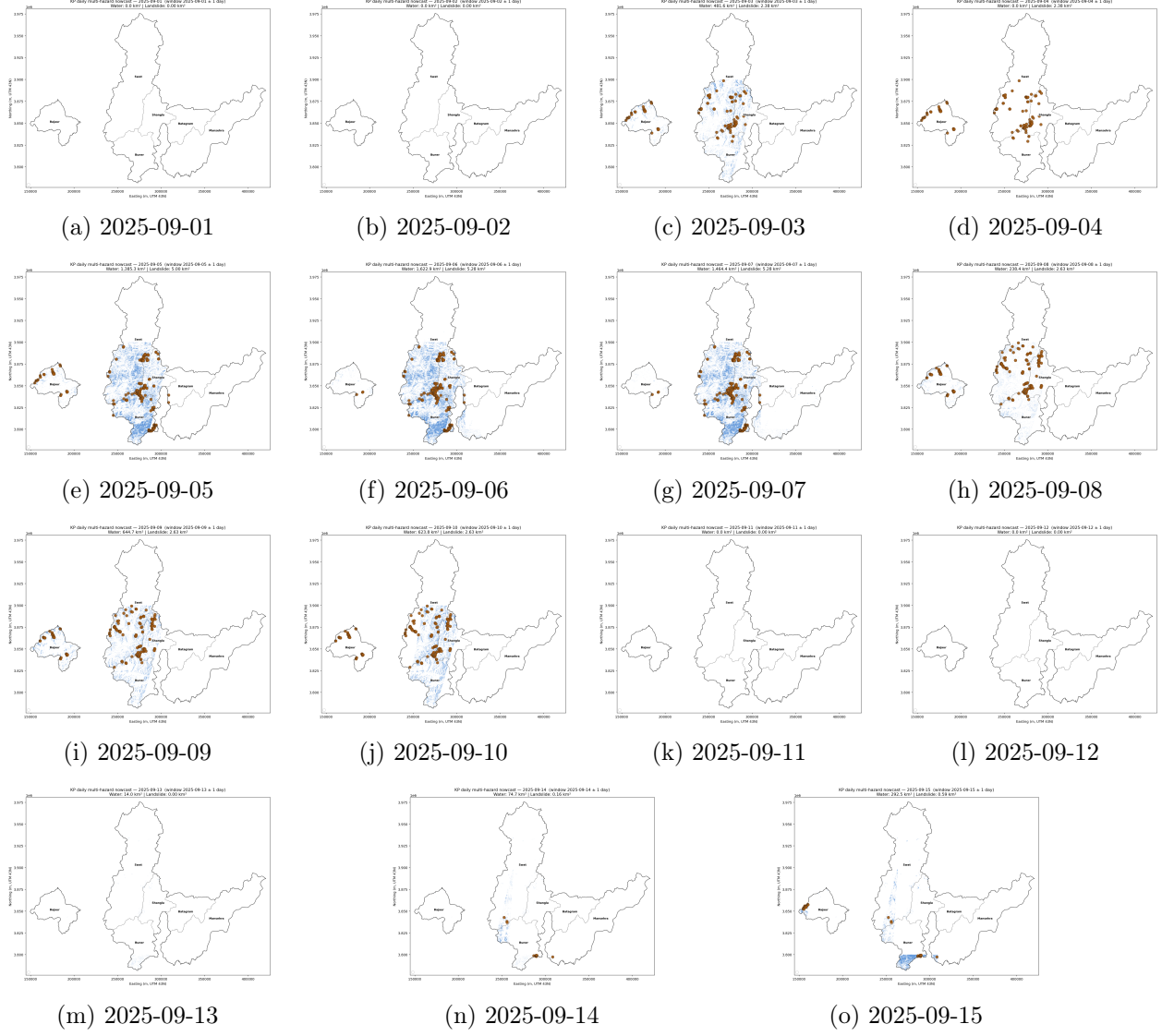


Figure 7: Multi-hazard maps for Khyber Pakhtunkhwa, September 2025 (days 1–15).

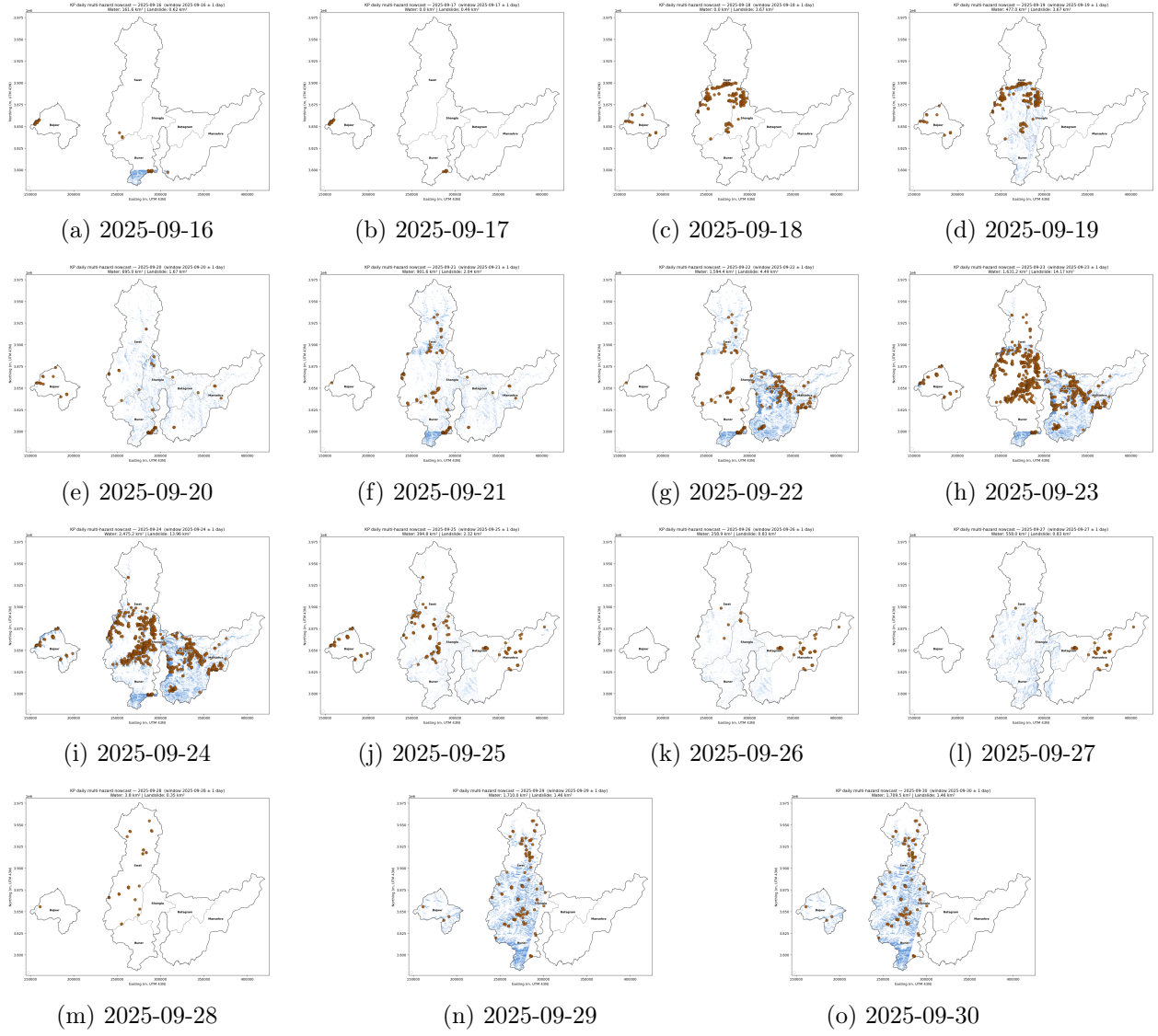


Figure 8: Multi-hazard maps for Khyber Pakhtunkhwa, September 2025 (days 16–30).

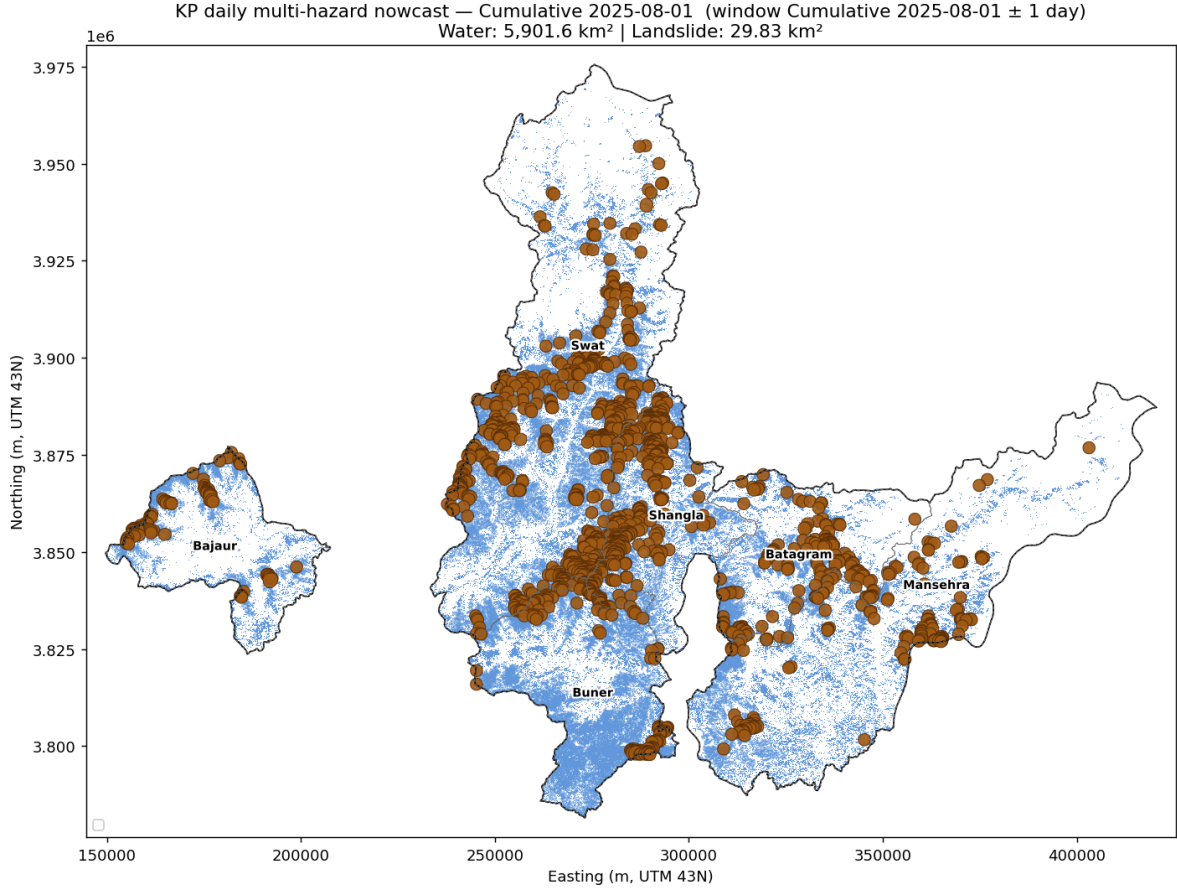


Figure 9: Cumulative multi-hazard footprint over Khyber Pakhtunkhwa for the 2025 monsoon study window (1 August–30 September 2025), obtained by aggregating the 60 daily hazard composites shown in the preceding figures. Each pixel represents the maximum hazard intensity recorded across the 60-day window, jointly integrating flash-flood, glacial lake outburst flood (GLOF), and rainfall-triggered landslide hazards. The map therefore delineates the spatial envelope of compound hydro-geomorphic hazard exposure during the peak monsoon period and provides the basis for the regional impact assessment discussed in the main text.